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TRABALHO DE CONCLUSÃO DE CURSO

APPLICATION OF THE DEEPLABCUT TOOLBOX IN FUTSAL

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RESUMO

Este estudo avaliou a aplicabilidade do DeepLabCut, uma ferramenta de *deep learning* baseada em *pose estimation*, para o rastreamento de jogadores e da bola em partidas de futsal 2v2. Utilizando imagens aéreas captadas por drone e a arquitetura ResNet-50, o modelo foi treinado com um conjunto de dados rotulado manualmente. A precisão foi avaliada com base no erro médio euclidiano entre as marcações manuais e as prediçõese, com bons resultados mesmo em condições reais de gravação. Transformação foi aplicada para corrigir distorções da filmagem, permitindo a análise em coordenadas métricas reais. Filtros baseados em probabilidade e limiares de movimentação refinaram os dados, excluindo predições incertas ou implausíveis. Embora limitações como perda de rastreamento em jogadores com pouca distinção de cor tenham sido observadas, os resultados demonstram a viabilidade do uso do DeepLabCut para análise de desempenho em esportes coletivos com recursos acessíveis e replicáveis.

PALAVRAS-CHAVE

Futsal. DeepLabCut. Computer Vision. Machine Learning. Motion Tracking.

ABSTRACT

This study evaluated the applicability of DeepLabCut, a deep learning-based pose estimation tool, for tracking players and the ball in 2v2 futsal matches. Aerial footage recorded by a drone was used, and the model was trained using the ResNet-50 architecture on a manually labelled dataset. Accuracy was assessed through the mean Euclidean error between manual annotations and network predictions, with satisfactory results despite real-world recording conditions. A perspective transformation was applied to correct geometric distortions in the footage, enabling analysis in real-world metric coordinates. Likelihood-based and range threshold filters were used to refine the dataset by removing uncertain or implausible predictions. Despite challenges such as tracking loss due to poor colour contrast, the findings demonstrate the feasibility of using DeepLabCut for performance and tactical analysis in team sports using affordable and replicable tools.

KEYWORDS

Futsal. DeepLabCut. Computer Vision. Machine Learning. Motion Tracking.

INTRODUCTION

Advances in artificial intelligence and machine learning have revolutionised the ability to detect and track dynamic objects and athletes in complex, fast-paced environments. Deep learning, with its capacity to process high-dimensional layered neural networks, has superseded traditional computer vision techniques, offering unprecedented accuracy in motion analysis. Recent works, such as imputing football player locations from sparse event data, e.g. shots and passes (Everett, Beal, Matthews, Early, Norman & Ramchurn, 2023), underscore the potential of these technologies. Similarly, the framework by Citraro, Márquez-Neila, Savarè et. al. (2020) for real-time pose estimation highlights the growing demand for precision in sports analytics.

DeepLabCut, an open-source deep learning toolkit, exemplifies these technological strides (Mathias & Mathis, 2019). Originally designed for animal pose estimation, it employs architectures such as ResNets and readout layers to track anatomical keypoints with high fidelity. By repurposing algorithms from human pose estimation, DeepLabCut has demonstrated versatility across domains, including sports science, where it enables analysis of athlete kinematics and tactical patterns. (Mathis et al., 2018).

However, applying such tools to team-based, high-velocity sports like futsal presents unique challenges. The sport's condensed playing area, rapid ball transitions, and frequent occlusions due to close player proximity demand tracking systems capable of examining fine-grained movements under dynamic conditions. This paper, following the work of Apolinario-Souza, Bedo and Leonardi (2024), evaluates DeepLabCut's efficacy in tracking players and the ball within a 2v2 futsal game.

METHODS

Online Material

All scripts, videos, and other materials used in this paper can be found at the following link: https://github.com/barthlucast/deeplabcut_futsal/tree/main

Dataset

Two pairs (four men aged 32±3.0 years) played in a small-sided game to maintain possession of the ball for as long as possible over 3 minutes. This experiment occurred on an open court within a delimited 6x6 meters. The data was recorded at 30Hz with a resolution of 640x480 pixels using a drone (DJI Mini SE) positioned at a height of 10 meters. The moments in the video where the ball left the designated area ('ball out of play') were removed, leaving 78 seconds remaining (online material: Original Video)

DeepLabCut provides a function that allows labelling all the extracted frames using an interactive graphical user interface (GUI). Using the GUI, we localised nine key points (the ball, the four players, and the four corners of the playing area) across 300 frames sampled using the k-means clustering approach within DeepLabCut (online material: Script 1)

Create a training dataset and train the network

We ran the model using the Google Colab GPU. To create the model, we employed the ResNet-50 architecture, a deep residual network known for its high performance in image recognition tasks. Additionally, we utilised the *imgaug* library for data augmentation, which enhances the variability of the training data by applying random transformations such as rotations, flips, changes in brightness, and contrast (online material: Script 2)

We trained the model using the function *deeplabcut.train_network*. This function initiates the training process for the neural network dedicated to pose estimation. To enhance the model's robustness, we employed the first shuffle of the dataset, providing a varied training set. Additionally, we set the function to save the model checkpoint every 500 iterations, ensuring that the training progress is preserved and enabling the possibility of resuming from the latest checkpoint in case of interruptions. Finally, we defined the training process to run for a maximum of 10000 iterations, preventing overfitting and ensuring efficient use of computation resources. We stopped the training when the loss function value reached below 0.007 with a learning rate (lr) of 0.005 (iteration: 4700; loss: 0.0068; lr: 0.005).

Evaluation of the trained network

We evaluated the performance of the trained network by calculating the mean average Euclidean error between the manual labels and those predicted by DeepLabCut. The model's results showed that the mean error in pixels (train error) during the model's training was 4.64, and during the evaluation on a test set (test error) was 3.03. The training error was 3.32 and the test error was 3.02, using a p-cutoff of 0.6.

Perspective transformation

Drone footage results in a video with fluctuations in the camera's position due to the wind's effects on the drone. We apply a perspective transformation to each frame to address the issue, using the four known fixed points in the image that delineate the playing area.

Our implementation computed the homography matrix and applied the perspective transformation (online material: Script 3). First, we extracted each frame's four corner points:

Bottom-left = (xA, yA)

Bottom-right = (xB, yB)

Top-right = (xC, yC)

Top-left = (xD, yD)

Then, we calculated the distances between these points to find the mean width and height:

$$\text{width}_{AD} = \sqrt{(x_A - x_D)^2 + (y_A - y_D)^2}$$
$$\text{width}_{BC} = \sqrt{(x_B - x_C)^2 + (y_B - y_C)^2}$$
$$\text{meanWidth} = \frac{\text{width}_{AD} + \text{width}_{BC}}{2}$$

$$\text{height}_{AB} = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2}$$
$$\text{height}_{CD} = \sqrt{(x_C - x_D)^2 + (y_C - y_D)^2}$$
$$\text{meanHeight} = \frac{\text{height}_{AB} + \text{height}_{CD}}{2}$$

Based on the known real-world distance, we derive the pixels-per-meter ratio:

$$\text{pixelsPerMeterHorizontal} = \frac{\text{meanWidth}}{\text{realDistance}}$$
$$\text{pixelsPerMeterVertical} = \frac{\text{meanHeight}}{\text{realDistance}}$$
$$\text{pixelsPerMeter} = \frac{\text{pixelsPerMeterHorizontal} + \text{pixelsPerMeterVertical}}{2}$$

Next, we define the desired coordinates in the new image using the pixels per meter ratio. Assume *d* is the actual distance between the cones:

$$\mathbf{pts\_dest} = \begin{bmatrix} 0 & 0 \\ d \cdot \text{pixels\_per\_meter} & 0 \\ d \cdot \text{pixels\_per\_meter} & d \cdot \text{pixels\_per\_meter} \\ 0 & d \cdot \text{pixels\_per\_meter} \end{bmatrix}$$

Here, pts_dest represents the coordinates in the destination plane, a transformed version of the original coordinates. Each point is represented in the format [x,y]:

1. [0, 0]: bottom-left corner;

2. [d · pixels_per_meter, 0]: bottom-right corner;

3. [d · pixels_per_meter, d · pixels_per_meter]: top-right corner;

4. [0, d · pixels_per_meter]: top-left corner.

We calculated the homography matrix *H* using the *cv2.getPerspectiveTransform* function of *OpenCV*:

H = cv2.getPerspectiveTransform(pts_orig, pts_dest)

For each point *p = [x, y]* of the elements, it is transformed using the homography matrix *H*:

$p' = H \cdot p$

where $p'=[x',y',1]$ is the transformed coordinate. After, we performed by multiplying the point vector by the homography matrix *H*:

$$[\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}]$$

Finally, we normalise the resulting vector by dividing it by the homogeneous coordinates:

$$[x' = \frac{h_{11}x+h_{12}y+h_{13}}{h_{31}x+h_{32}y+h_{33}} \quad , \quad y' = \frac{h_{21}x+h_{22}y+h_{23}}{h_{31}x+h_{32}y+h_{33}}]$$

The filter with the likelihood

After performing the perspective transformation of the data, we filtered the x and y position values based on the probability. Based on their likelihood values, the equation replaces values with 'NaN' (Not a Number) in the x and y position rows. Given a matrix E ∈ R3×n where E represents the positions (E1= position x, E2=position y) and likelihoods (E3) of elements, as follows:

$$M_j = \begin{cases} 1 & \text{if } E_{3,j} < 0.80 \\ 0 & \text{otherwise} \end{cases} \quad \text{for } j = 1, 2, \ldots, n$$

$$E_{1,j} = \begin{cases} \text{NaN} & \text{if } M_j = 1 \\ E_{1,j} & \text{otherwise} \end{cases} \quad \text{for } j = 1, 2, \ldots, n$$

$$E_{2,j} = \begin{cases} \text{NaN} & \text{if } M_j = 1 \\ E_{2,j} & \text{otherwise} \end{cases} \quad \text{for } j = 1, 2, \ldots, n$$

We utilised probability values greater than 0.80.

The filter with the threshold limits

We established two threshold limits: an upper limit of 8 and a lower limit of -2. The filtering process then replaced any values in the dataset that exceeded the upper limit or fell below the lower limit with NaN. We subsequently added the filtered data to the d matrix. This approach excludes data points outside the defined range, improving the dataset's accuracy and relevance for further analysis. The filtered data matrix dfiltered is obtained by:

$$\mathbf{d}_{\text{filtered}} = \begin{cases} \begin{bmatrix} \text{NaN} \\ \text{NaN} \end{bmatrix} & \text{if } x > \text{lim}_{\text{up}} \text{ or } y > \text{lim}_{\text{up}} \text{ or } x < \text{lim}_{\text{down}} \text{ or } y < \text{lim}_{\text{down}} \\ \begin{bmatrix} x \\ y \end{bmatrix} & \text{otherwise} \end{cases}$$

The tracking performance for the player in the black shirt was notably suboptimal, likely attributable to challenges posed by outdoor recording conditions. The grey playing field and dynamic shadows reduced colour contrast, causing the black jersey to blend into the background. The same was expected to happen to the originally white ball, which was therefore coloured pink in Adobe Premiere. These environmental factors led to frequent dropouts for the black shirt player, where tracking failures were more pronounced, suggesting the colour-related artefacts significantly impair tracking accuracy for specific targets.

RESULTS

The performance of the DeepLabCut model, trained using the ResNet-50 architecture and optimised over 4,700 iterations, showed high accuracy in tracking both players and the ball in a small-sided futsal environment. The model achieved a training error of 4.64 pixels and a test error of 3.03 pixels. When applying a confidence (likelihood) threshold of 0.6 to enhance prediction reliability, the training error decreased to 3.32 pixels, and the test error improved slightly to 3.02 pixels. These results indicate that the model effectively generalised to previously unseen data despite the limited dataset size.

The application of a perspective transformation, based on a homography matrix calculated from the four known corners of the playing area, allowed the conversion of the original video feed, subject to drone instability and optical distortion, into a stable, top-down orthographic projection. This transformation enabled accurate mapping of player and ball positions into real-world coordinates, which is essential for any subsequent kinematic or tactical analysis.

Following this, two filtering steps were applied to the positional data. The likelihood filter, which removed coordinate estimates with a confidence score below 0.80, served to eliminate low-certainty predictions. The range threshold filter excluded position values beyond a realistic movement boundary (set between -2 and 8 meters), which further refined the dataset by eliminating outliers potentially caused by occlusions or abrupt movements.

However, some limitations in detection quality were observed. Notably, the player wearing a black shirt exhibited a higher rate of tracking dropout. This was likely due to low colour contrast between the dark clothing and the grey court, as well as dynamic shadows from outdoor lighting. In contrast, the ball, visually modified to pink in post-processing, was consistently tracked, demonstrating how visual adjustments can significantly impact tracking reliability.

Overall, the system was capable of generating temporally continuous and spatially accurate tracking data under real-world constraints, laying the groundwork for more advanced player behaviour and tactical analysis.

DISCUSSION

This study evaluated the feasibility of using DeepLabCut, an open-source pose estimation framework, for tracking players and the ball in a small-sided futsal scenario recorded under real-world conditions using a drone. The results demonstrate that DeepLabCut, when properly configured and trained, can yield accurate and consistent tracking data suitable for kinematic and tactical analyses in sports settings.

The application of a homography-based perspective transformation played a central role in addressing geometric distortion inherent to aerial footage. By identifying four known fixed points on the court (the corners of the playing area), we computed a homography matrix to project each frame into a normalised plane. This transformation enabled the conversion of raw pixel coordinates into a consistent metric reference frame, allowing movement trajectories to be interpreted within real-world distances (meters), which is essential for meaningful tactical or biomechanical analysis.

Complementing the spatial correction, two filtering procedures were implemented to ensure data integrity. First, a likelihood-based filter removed predicted keypoints with low confidence scores (below 0.80). This threshold, derived from DeepLabCut's output probabilities, effectively discarded uncertain predictions that would have introduced noise into the positional data. Second, a range-based threshold filter excluded position values outside the physical boundaries of the court, based on plausible motion constraints for futsal. Together, these filters eliminated both statistical outliers and implausible trajectories, substantially improving the dataset's usability.

Notably, this post-processing pipeline also revealed key insights into environmental challenges. The player wearing a black jersey experienced frequent tracking dropouts, particularly in frames with heavy shadowing. The lack of colour contrast between the jersey and the court background, combined with variable lighting conditions, impaired model confidence, even before filtering. In contrast, the ball, originally white but recoloured pink in post-production, was tracked with high consistency. This finding suggests that visual preprocessing (e.g., colour manipulation or contrast enhancement) can be a critical step in improving downstream pose estimation performance, particularly in uncontrolled or outdoor environments. Moreover, when feasible, such visual adjustments should be planned prior to data collection to mitigate known limitations associated with lighting and contrast variability.

Another key conclusion is that the fidelity of a sports tracking system depends not only on model accuracy but also on how effectively raw predictions are filtered, corrected, and adapted to the contextual demands of the sport under study. In our case, it was only through a sequence of post-processing steps that the raw DeepLabCut output became actionable data.

In practical terms, this study offers a blueprint for implementing deep learning-based tracking in sports science, providing a viable and cost-effective alternative. By leveraging widely available tools (e.g., Google Colab, OpenCV) and open-source software, researchers and practitioners can generate accurate spatiotemporal data from single-camera recordings, even in field conditions. Importantly, this requires designing a thoughtful post-processing workflow.

CONCLUSION

To summarise, this study affirms the potential of DeepLabCut for automated motion analysis in futsal and similar team sports. When combined with appropriate pre- and post-processing techniques, the tool provides accurate, scalable, and reproducible tracking that can support a range of applications, from performance diagnostics to tactical evaluation. Continued methodological refinements and attention to visual design will be essential for maximising the effectiveness of deep learning-based tracking in real-world sporting contexts.

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